

From the scaling hypothesis to the Renormalization Group

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The scaling hypothesis: mean field

From the saddle-point equation for uniform external field

$$r_0 t M + u_0 M^3 - h = 0 \quad (1)$$

by rescaling variables as $M = \tilde{M}|r_0 t|^{1/2}$ and $h = \tilde{h}|r_0 t|^{3/2}$,

$$\implies \tilde{h} = \frac{r_0 t}{|r_0 t|} \tilde{M} + u_0 \tilde{M}^3. \quad (2)$$

which leads to a scaling form for the magnetization

$$M(t, h) = |r_0 t|^{\frac{1}{2}} \left(\frac{h - u_0 M^3}{|r_0 t|^{3/2}} \right) \frac{r_0 t}{|r_0 t|} \quad (3)$$

$$\approx |r_0 t|^{\frac{1}{2}} \left(\frac{h}{|r_0 t|^{3/2}} \right) \frac{r_0 t}{|r_0 t|}. \quad (4)$$

$$\implies M(t, h) = |t|^{\frac{1}{2}} g_M^{\pm} \left(\frac{h}{|t|^{\frac{3}{2}}} \right). \quad (5)$$

The scaling hypothesis: mean field

The saddle-point free energy is

$$f(t, h) = \min_m \left[\frac{r_0 t}{2} m^2 + \frac{u_0}{4} m^4 - h m \right] = \begin{cases} -\frac{r_0^2 t^2}{4 u_0} & h = 0, t < 0 \\ -\frac{3}{4} h^{4/3} u_0^{-1/3} & h \neq 0, t = 0 \end{cases} \quad (6)$$

using $M = (h/u_0)^{1/3}$ for $t = 0$.

This expression can be written in a scaling form as follows

$$f(t, h) = |r_0 t|^2 \begin{cases} -\frac{1}{4 u_0} & h = 0, t < 0 \\ -\frac{3}{4} \frac{h^{4/3}}{r_0^2 t^2 u_0^{1/3}} & h \neq 0, t = 0 \end{cases} \quad (7)$$

$$= |t|^2 g_f \left(\frac{h}{|t|^{\frac{3}{2}}} \right) \quad (8)$$

with $g_f(x) \sim \text{const.}$ for $x \rightarrow 0$ and $g_f(x) \sim x^{4/3}$ for $x \rightarrow \infty$.

The scaling hypothesis: beyond mean field

A scaling behavior of the free energy close to criticality should be correct beyond mean field,

$$f_{\text{sing}}(h, t) \sim |t|^{2-\alpha} g_f \left(\frac{h}{|t|^\Delta} \right), \quad (9)$$

where

- exponents α and Δ (*gap exponent*) are not known in general,
- scaling function $g_f(x) \sim \begin{cases} \text{const.} & x \rightarrow 0 \\ x^\epsilon & x \rightarrow \infty \end{cases}$

Plan: derive all relevant physical quantities from f_{sing} and check whether critical exponents satisfy some *scaling relations*.

The scaling hypothesis: beyond mean field

Magnetization: from $M \sim \partial f / \partial h$, close to criticality

$$M \sim |t|^{2-\alpha-\Delta} g'_f \left(\frac{h}{|t|^\Delta} \right), \quad (10)$$

Since $M \sim |t|^\beta$, we obtain the relation $2 - \alpha - \Delta = \beta$.

Susceptibility: from $\chi \sim \partial M / \partial h$, close to criticality

$$\chi \sim |t|^{2-\alpha-2\Delta} g''_f \left(\frac{h}{|t|^\Delta} \right), \quad (11)$$

Since $\chi \sim |t|^{-\gamma}$, we obtain relation $-\gamma = 2 - \alpha - 2\Delta$.

Eliminating Δ from the two gives the scaling relation

$$\alpha + 2\beta + \gamma = 2 \quad (12)$$

known as *Rushbrooke's law*.

The scaling hypothesis: beyond mean field

The **critical isothermal** $M \sim h^{1/\delta}$ should be also obtained as

$$\lim_{|t| \rightarrow 0} |t|^{2-\alpha-\Delta} \left. \frac{\partial}{\partial x} g_f(x) \right|_{x=h|t|^{-\Delta}} \sim |t|^{2-\alpha-\Delta} \left. \frac{\partial}{\partial x} x^\epsilon \right|_{x=h|t|^{-\Delta}} \quad (13)$$

$$\sim |t|^{2-\alpha-\Delta} h^{\epsilon-1} |t|^{-\Delta(\epsilon-1)} \quad (14)$$

$$\sim h^{\epsilon-1} |t|^{2-\alpha-\epsilon\Delta} \quad (15)$$

$$\sim h^{\epsilon-1} |t|^{\beta+(1-\epsilon)\Delta} \quad (16)$$

which implies $\epsilon - 1 = 1/\delta$, i.e. $\epsilon = (\delta + 1)/\delta$, and

$$\Delta = \beta\delta. \quad (17)$$

Combining this with previous scaling relation gives

$$-\gamma = 2 - \alpha - 2\Delta = 2 - (2 - 2\beta - \gamma) - 2\beta\delta \quad (18)$$

$$\implies \gamma = (\delta - 1)\beta \quad (19)$$

known as **Griffith's law**.

The scaling hypothesis: beyond mean field

From the phenomenological scaling ansatz for the **correlation function**

$$G(\vec{r}) \sim \frac{1}{|\vec{r}|^{d-2+\eta}} Y\left(\frac{\vec{r}}{\xi}\right), \quad \eta \geq 0, \quad (20)$$

integrating over the whole space,

$$\begin{aligned} \chi_T &\sim \int d\vec{r} \frac{1}{|\vec{r}|^{d-2+\eta}} Y\left(\frac{\vec{r}}{\xi}\right) \\ &\approx \int d\vec{z} \frac{\xi^d}{|z|^{d-2+\eta}} \xi^{2-d-\eta} Y(z) \\ &\approx \xi^{2-\eta} \int d\vec{z} \frac{1}{|z|^{d-2+\eta}} Y(z) \\ &\propto |t|^{-(2-\eta)\nu}, \end{aligned}$$

which compared with $\chi \sim |t|^{-\gamma}$ gives **Fisher's law**

$$\gamma = (2 - \eta) \nu. \quad (21)$$

The scaling hypothesis: beyond mean field

The general dimensional scaling of the free energy close to criticality is

$$F = \left(\frac{L}{\xi}\right)^d \tilde{f}_\xi + \cdots + \left(\frac{L}{a}\right)^d \tilde{f}_a, \quad (22)$$

where $\tilde{f}_\xi, \dots, \tilde{f}_a$ are regular parts of the free energy.

If ξ is the only relevant length scale at criticality (for $d < d_{uc}$),

$$F \approx \left(\frac{L}{\xi}\right)^d \tilde{f}_\xi \implies f_{\text{sing}} \sim \xi^{-d} \sim |t|^{d\nu}. \quad (23)$$

But $f_{\text{sing}} \sim |t|^{2-\alpha}$, so we find (for non mean-field systems)

$$d\nu = 2 - \alpha. \quad (24)$$

known as ***Josephson's hyperscaling law***.

The Renormalization Group: basic concepts

Scaling theory states that sufficiently close to the critical point:

- correlation length ξ is the only relevant length scale,
- statistical scale invariance is valid up to that length scale.

If scaling theory is correct, the critical properties of a physical system described by an action $S[\phi]$ for a spatial field $\phi(\vec{x})$ should not explicitly depend on degrees of freedom at distances $x \ll \xi$.

So, why don't we eliminate such dependence?

This can be correctly done with a procedure known as the ***Renormalization Group (RG) transformation***.

The Renormalization Group: basic concepts

The RG steps are:

- 1 COARSE-GRAINING: trace out all degrees of freedom from Λ^{-1} to ℓ/Λ with $\ell > 1$ in order to obtain a coarse-grained field

$$\tilde{\phi}(\vec{x}) = \frac{1}{\ell^d} \int_{\mathcal{B}_\ell(\vec{x})} d\vec{x}' \phi(\vec{x}'). \quad (25)$$

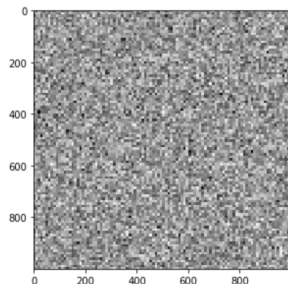
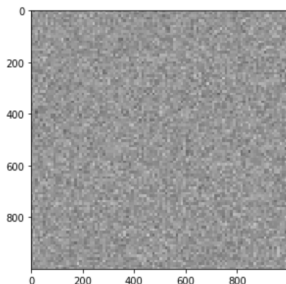
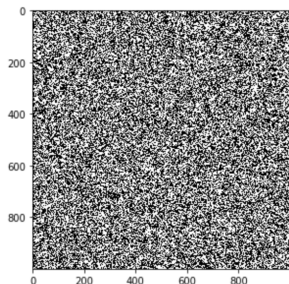
- 2 RESCALING: rescale length scales as $\vec{x} \rightarrow \vec{x}' = \vec{x}/\ell$ in order to restore the original resolution.
- 3 FIELD RENORMALIZATION: restore the correct scale of fluctuations multiplying by a non-dimensional constant ζ , i.e.

$$\phi'(\vec{x}') = \frac{1}{\zeta} \tilde{\phi}(\vec{x}). \quad (26)$$

After these steps are performed we get a **renormalized action** $S'[\phi']$.

The Renormalization Group: basic concepts

EXAMPLE: Independent and uniformly random binary points on a square lattice (I). Coarse graining reduces resolution which is restored by spatial rescaling (II). Amplitude rescaling restores contrast (III).



The Renormalization Group: basic concepts

Since only short wavelengths are affected, at criticality, f_{sing} (and then $S[\phi]$) should be functionally invariant under RG.

Original action:

$$S[\phi] = \int d\vec{x} \left[\frac{rt}{2} \phi^2 + \frac{u}{4} \phi^4 + \frac{v}{6} \phi^6 + \dots + \frac{c}{2} (\nabla \phi)^2 + \frac{c_4}{2} (\nabla^2 \phi)^2 + \dots \right] \quad (27)$$

Action after the RG transformation

$$S'[\phi'] = \text{const.} + \int d\vec{x}' \left[\frac{rt'}{2} \phi'^2 + \frac{u'}{4} \phi'^4 + \frac{v'}{6} \phi'^6 + \dots + \frac{c'}{2} (\nabla \phi')^2 + \frac{c'_4}{2} (\nabla^2 \phi')^2 + \dots \right] \quad (28)$$

The parameters appearing in the theory got **renormalised**.

The Renormalization Group: basic concepts

The action corresponds to a point $\mathcal{P} = (t, u, v, \dots, c, c_4, \dots)$ in a (possibly infinite dimensional) parameter space which undergoes a flow described by the RG mapping

$$\mathcal{R}_\ell \mathcal{P} = \mathcal{P}' \quad (29)$$

More precisely, the RG transformation defines a *semigroup* with

$$\mathcal{R}_1 = \mathbb{I}, \quad \mathcal{R}_{\ell_1} \mathcal{R}_{\ell_2} = \mathcal{R}_{\ell_1 \ell_2} \quad \forall \ell_1, \ell_2 \geq 1 \quad (30)$$

REMARK: It is not a group because there is no inverse since $\ell \geq 1$.

The RG transformation is based on a *dilation* operation, whose fixed points are ***scale-invariant***. Physical quantities should become homogeneous close to a scale-invariant point.

The Renormalization Group: basic concepts

Linearizing the RG transformation close to a fixed point $\mathcal{P}^*$,

$$\mathcal{P}_\alpha \simeq \mathcal{P}_\alpha^* + \delta\mathcal{P}_\alpha = \mathcal{P}_\alpha^* + \sum_\beta \frac{\partial \mathcal{P}_\alpha}{\partial \mathcal{P}_\beta} \delta\mathcal{P}_\beta, \quad (31)$$

where $\{\alpha\}$ is the parameter set.

The matrix

$$(R_\ell)_{\alpha\beta} = \frac{\partial \mathcal{P}_\alpha}{\partial \mathcal{P}_\beta} \quad (32)$$

is the generator of the infinitesimal RG transformations, for which it exists a set of eigenvalues $\{\lambda_i(\ell)\}$ and a set of eigenvectors $\{v_i\}$, such that

$$R_\ell v_i = \lambda_i(\ell) v_i. \quad (33)$$

The Renormalization Group: basic concepts

Because of the semigroup property,

$$\lambda_i(\ell_1)\lambda_i(\ell_2)v_i = \lambda_i(\ell_1\ell_2)v_i \quad (34)$$

together with $\lambda_i(1) = 1$ implies that $\lambda_i(\ell) = \ell^{y_i}$, where y_i is the ***anomalous dimension*** (associated to $\mathcal{P}^*$) along the eigen-direction v_i .

Assuming the eigen-decomposition $\mathcal{P}' = \mathcal{P}^* + \sum_i \ell^{y_i} g_i v_i$, then

- if $y_i > 0$, the *coupling* (parameter) g_i grows under RG scaling and v_i is a ***relevant*** direction;
- if $y_i < 0$, then g_i decreases under RG scaling and v_i is an ***irrelevant*** direction;
- if $y_i = 0$, then v_i is a ***marginal*** direction and the behavior is determined by non-linear terms.

The Renormalization Group: basic concepts

If a physical observable X at a scale-invariant point is a homogeneous function of the physical parameters $\{g_i\}$, i.e.

$$X(g_1, g_2, \dots) = \ell^{y_X} X(\ell^{y_{g_1}} g_1, \ell^{y_{g_2}} g_2, \dots), \quad (35)$$

where y_X is the scaling dimension of X .

The expression is redundant, so we can set $\ell^{y_{g_1}} g_1 = 1$, and use it to eliminate ℓ by replacing $\ell = g_1^{-1/y_{g_1}}$,

$$X(g_1, g_2, \dots) = g_1^{-y_X/y_{g_1}} X\left(1, \frac{g_2}{g_1^{y_{g_2}/y_{g_1}}}, \dots\right), \quad (36)$$

where y_{g_2}/y_{g_1} can be viewed as a **crossover exponent** determining the role of g_2 compared to g_1 .

Scaling from renormalization

The partition function should be invariant under RG, i.e.

$$e^{-\beta V f(t,h)} = Z = e^{-\beta V' f(t',h')} \quad (37)$$

$$\implies f(t, h) = \ell^{-d} f(t', h') = \ell^{-d} f(\ell^{y_t} t, \ell^{y_h} h) \quad (38)$$

with scaling dimension $y_f = -d$.

Setting $\ell^{y_t} t = 1$,

$$f(t, h) = t^{\frac{d}{y_t}} f\left(1, t^{-\frac{y_h}{y_t}} h\right) \quad (39)$$

$$= t^{\frac{d}{y_t}} g_f\left(\frac{h}{t^{y_h/y_t}}\right) \quad (40)$$

From previous results, we identify $2 - \alpha = d/y_t$ and $\Delta = y_h/y_t$.

Scaling from renormalization

The order parameter

$$M(t, h) = -\frac{1}{\beta V} \frac{\partial}{\partial h} \log Z(t, h) \quad (41)$$

$$= -\frac{\ell^{y_h}}{\ell^d V'} \frac{\partial}{\beta \partial h'} \log Z'(t', h') \quad (42)$$

$$= \ell^{y_h - d} M(\ell^{y_t} t, \ell^{y_h} h), \quad (43)$$

$$= t^{\frac{d - y_h}{y_t}} M(1, h t^{-y_h/y_t}) \quad (44)$$

which gives $\beta = (d - y_h)/y_t$.

The correlation length scales as

$$\xi(t, h) = \ell \xi(t', h') \quad (45)$$

$$= \ell \xi(\ell^{y_t} t, \ell^{y_h} h) \quad (46)$$

$$= t^{-\frac{1}{y_t}} \xi(1, h t^{-y_h/y_t}) \quad (47)$$

which gives $\nu = 1/y_t$ and thus the scaling relation $2 - \alpha = d\nu$.

Scaling from renormalization

The two-point correlation function $G(\vec{x} - \vec{x}')$ transforms as follows

$$G(r, t, h) = \ell^{2(y_h - d)} G(r/\ell, \ell^{y_t} t, \ell^{y_h} h) \quad (48)$$

$$= t^{2(d - y_h)/y_t} G(rt^{1/y_t}, 1, ht^{-y_h/y_t}) \quad (49)$$

$$= \frac{1}{r^{2(d - y_h)}} g_G(rt^{1/y_t}, ht^{-y_h/y_t}), \quad (50)$$

where $[G] = [M]^2$ and a non-dimensional scaling function g_G . Compared to previous results, it must be

$$2(d - y_h) = d - 2 + \eta. \quad (51)$$

REMARK: From MF theory, at criticality, $G(r) \propto r^{2-d}/c$. To ensure $\eta > 0$, the parameter c should scale in a non-trivial way under RG.

Scaling properties of the Gaussian model

Consider the Fourier representation of the Gaussian model,

$$S_G[\hat{\phi}] = -\frac{1}{2V} \sum_{\vec{k}} (\bar{c}k^2 + \bar{r}t) |\hat{\phi}(\vec{k})|^2 - \frac{1}{V} \sum_{\vec{k}} h(\vec{k}) \hat{\phi}(-\vec{k}), \quad (52)$$

The partition function is computed explicitly by Gaussian integration,

$$Z_G[h] \propto \exp \left[\frac{1}{2} \sum_{\vec{k}}^{|\vec{k}| < \Lambda} \log \frac{2\pi}{ck^2 + rt} + \frac{\beta}{2V} \sum_{\vec{k}}^{|\vec{k}| < \Lambda} \frac{\hat{h}(\vec{k}) \hat{h}(-\vec{k})}{\bar{c}k^2 + \bar{r}t} \right] \quad (53)$$

$$= e^{\frac{\beta h^2 V}{2\bar{r}t}} \prod_{\vec{k}} \left(\frac{2\pi}{rt + ck^2} \right)^{\frac{1}{2}} \quad (54)$$

Scaling properties of the Gaussian model

The free energy density in the presence of uniform external field h is

$$f(t, h) = -\frac{k_B T}{2} \int \frac{d\vec{k}}{(2\pi)^d} \log \frac{2\pi}{rt + ck^2} - \frac{h^2}{2\bar{r}t} \quad (55)$$

$$\begin{aligned} &= -\frac{k_B T}{2} \frac{N}{V} \log 2\pi \\ &\quad + \frac{k_B T}{2} \int \frac{d\vec{k}}{(2\pi)^d} \log (rt + ck^2) - \frac{h^2}{2\bar{r}t} \end{aligned} \quad (56)$$

$$= f_0 + f_{\text{sing}}(t, h), \quad (57)$$

with

$$f_{\text{sing}}(t, h) = \frac{k_B T}{2} \int^\Lambda dk k^{d-1} \int_{\mathbb{S}^{d-1}} d\Omega \log (rt + ck^2) - \frac{h^2}{2\bar{r}t}.$$

Scaling properties of the Gaussian model

Noticing that the angular integral can be performed in terms of

$K_d = S_d/(2\pi)^d$ and $S_d = \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})}$, i.e. the surface of the $(d-1)$ -dim sphere $\mathbb{S}^{d-1}$ embedded into a d -dimensional Euclidean space, we obtain

$$f_{\text{sing}}(t, h) = \frac{K_d}{2\beta} \int^\Lambda dk k^{d-1} \log(rt + ck^2) - \frac{h^2}{2rt} \quad (58)$$

$$= \frac{K_d}{2\beta} \left(\frac{rt}{c}\right)^{\frac{d}{2}} \left[\int_0^{\sqrt{\frac{c}{rt}}} dx x^{d-1} [\log rt + \log(1+x^2)] - \frac{h^2}{(rt)^{1+d/2}} \frac{c^{d/2}}{K_d} \right] \quad (59)$$

$$= t^{2-\alpha} g_f\left(\frac{h}{t^\Delta}\right) \quad (60)$$

with the critical exponents $\alpha = 2 - \frac{d}{2}$ and $\Delta = \frac{1}{2} + \frac{d}{4}$.

Wilson's Momentum Shell RG

Wilson's Idea: perform RG in momentum space integrating over Fourier modes in the range $\Lambda/\ell < |\vec{k}| < \Lambda$.

Let us decompose $\hat{\phi}(\vec{k}) = \hat{\phi}_>(\vec{k}) + \hat{\phi}_<(\vec{k})$ defined as:

- **fast modes** (high-energy)

$$\hat{\phi}_>(\vec{k}) = \begin{cases} \hat{\phi}(\vec{k}) & \frac{\Lambda}{\ell} < |\vec{k}| < \Lambda \\ 0 & \text{otherwise} \end{cases} \quad (61)$$

- **slow modes** (low-energy)

$$\hat{\phi}_<(\vec{k}) = \begin{cases} \hat{\phi}(\vec{k}) & |\vec{k}| < \frac{\Lambda}{\ell} \\ 0 & \text{otherwise.} \end{cases} \quad (62)$$

A generic action can be decomposed as follows

$$S \left[\left\{ \hat{\phi}(\vec{k}) \right\} \right] = S_0 \left[\left\{ \hat{\phi}_<(\vec{k}) \right\} \right] + S_0 \left[\left\{ \hat{\phi}_>(\vec{k}) \right\} \right] + S_I \left[\left\{ \hat{\phi}_>(\vec{k}), \hat{\phi}_<(\vec{k}) \right\} \right] \quad (63)$$

Wilson's Momentum Shell RG

STEP 1: Integrate over fast modes,

$$Z = \int \prod_{|\vec{k}| < \Lambda} d\hat{\phi} e^{-S[\{\hat{\phi}(\vec{k})\}]} \quad (64)$$

$$\begin{aligned} &= \int \prod_{|\vec{k}| < \Lambda/\ell} d\hat{\phi}_{<} e^{-S_0[\{\hat{\phi}_{<}(\vec{k})\}]} \int \prod_{\Lambda/\ell < |\vec{k}| < \Lambda} d\hat{\phi}_{>} e^{-S_0[\{\hat{\phi}_{>}(\vec{k})\}] - S_I[\{\hat{\phi}_{>}(\vec{k}), \hat{\phi}_{<}(\vec{k})\}]} \\ &= \int \prod_{|\vec{k}| < \Lambda/\ell} d\hat{\phi}_{<} e^{-S'[\{\hat{\phi}_{<}(\vec{k})\}]}, \end{aligned} \quad (65)$$

where we defined a **coarse-grained action**

$$\begin{aligned} S'[\{\hat{\phi}_{<}(\vec{k})\}] &= S_0[\{\hat{\phi}_{<}(\vec{k})\}] \\ &\quad - \log \left\{ \int \prod_{\Lambda/\ell < |\vec{k}| < \Lambda} d\hat{\phi}_{>} e^{-S_0[\{\hat{\phi}_{>}(\vec{k})\}] - S_I[\{\hat{\phi}_{>}(\vec{k}), \hat{\phi}_{<}(\vec{k})\}]} \right\} \end{aligned} \quad (66)$$

Wilson's Momentum Shell RG

STEP 2: Rescale momenta introducing $\vec{k}' = \vec{k}\ell$,

$$Z = \int \prod_{|\vec{k}'| < \Lambda} d\hat{\phi}_{<} e^{-S'[\{\hat{\phi}_{<}(\vec{k}')\}]},$$

STEP 3: Introduce a renormalized field $\hat{\phi}_{<}(\vec{k}') = z^{-1}\hat{\phi}'(\vec{k}')$,

$$Z = \int \prod_{|\vec{k}'| < \Lambda} d\hat{\phi}' e^{-S'[\{\hat{\phi}'(\vec{k}')\}]},$$

in which the renormalized action $S'[\hat{\phi}']$ contains a renormalized set of parameters, which are functions of (1) the original (“bare”) parameters, (2) the scale factor ℓ , (3) the constant z (related to ζ).

Next step: determining z and obtain RG flow equations for the parameters.

Wilson's Momentum Shell RG: Gaussian model

From the action of the Gaussian model in Fourier space,

$$S_G = \int^\Lambda \frac{d\vec{k}}{(2\pi)^d} \frac{ck^2 + rt}{2} \hat{\phi}(\vec{k}) \hat{\phi}(-\vec{k}) \quad (67)$$

since Fourier modes are independent,

$$S_G[\hat{\phi}] = S_G[\hat{\phi}_< + \hat{\phi}_>] = S_G[\hat{\phi}_<] + S_G[\hat{\phi}_>] \quad (68)$$

thus the coarse-graining step can be carried out exactly,

$$Z_G = \int D\hat{\phi}_< \int D\hat{\phi}_> e^{-S_G[\hat{\phi}_< + \hat{\phi}_>]} \quad (69)$$

$$= \int D\hat{\phi}_< e^{-S_G[\hat{\phi}_<]} \int D\hat{\phi}_> e^{-S_G[\hat{\phi}_>]} \quad (70)$$

$$\propto \int D\hat{\phi}_< e^{-S_G[\hat{\phi}_<]}. \quad (71)$$

Wilson's Momentum Shell RG: Gaussian model

The **coarse-grained** action has the same functional form as the original one,

$$S_G[\hat{\phi}_<] = \int^{\Lambda/\ell} \frac{d\vec{k}}{(2\pi)^d} \frac{ck^2 + rt}{2} \hat{\phi}_<(\vec{k}) \hat{\phi}_<(-\vec{k}) \quad (72)$$

Then **rescaling** momenta $\vec{k}' = \vec{k}\ell$,

$$S'_G[\hat{\phi}_<] = \int^{\Lambda} \frac{d\vec{k}'}{(2\pi)^d} \frac{\ell^{-d}}{2} (ck'^2\ell^{-2} + rt) \hat{\phi}_<(\vec{k}') \hat{\phi}_<(-\vec{k}'). \quad (73)$$

and introducing **field renormalization** $\hat{\phi}'(\vec{k}') = z^{-1} \hat{\phi}_<(\vec{k})$,

$$S'_G[\hat{\phi}'] = \int^{\Lambda} \frac{d\vec{k}'}{(2\pi)^d} \frac{\ell^{-d}}{2} (ck'^2\ell^{-2} + rt) z^2 \hat{\phi}'(\vec{k}') \hat{\phi}'(-\vec{k}'). \quad (74)$$

Wilson's Momentum Shell RG: Gaussian model

Setting $z = \ell^w$,

$$S'_G[\hat{\phi}'] = \int^\Lambda \frac{d\vec{k}}{(2\pi)^d} \frac{\ell^{-d}}{2} (c k^2 \ell^{-2} + r t) \ell^{2w} \hat{\phi}'(\vec{k}) \hat{\phi}'(-\vec{k}) \quad (75)$$

$$= \int^\Lambda \frac{d\vec{k}}{(2\pi)^d} \frac{1}{2} (c k^2 \ell^{-2+2w-d} + r t \ell^{2w-d}) \hat{\phi}'(\vec{k}) \hat{\phi}'(-\vec{k}). \quad (76)$$

$$= \int^\Lambda \frac{d\vec{k}}{(2\pi)^d} \frac{1}{2} (c'(\ell) k^2 + r t'(\ell)) \hat{\phi}'(\vec{k}) \hat{\phi}'(-\vec{k}), \quad (77)$$

we obtain the original form for $S'_G[\hat{\phi}']$, with renormalized parameters

$$c'(\ell) = c \ell^{-2+2w-d} \quad (78)$$

$$t'(\ell) = t \ell^{2w-d}. \quad (79)$$

The choice $w = 1 + d/2$ ensures $c' = c$ (**canonical scaling**), and

$$t'(\ell) = t \ell^2, \quad (80)$$

The RG flow has fixed points at $t = 0$ and $t = +\infty$.

Wilson's Momentum Shell RG: Gaussian model

EXERCISE: Consider the more general quadratic model, with external field and high-order transport term,

$$S_G[\phi] = \int d\vec{x} \left[\frac{c}{2} (\nabla \phi)^2 + \frac{c_4}{2} (\nabla^2 \phi)^2 + \frac{rt}{2} \phi^2 - h\phi \right]. \quad (81)$$

Repeating Wilson's RG procedure obtain a functionally equivalent renormalized action with the renormalized parameters

$$\begin{cases} t'(\ell) &= z^2 \ell^{-d} t \\ c'(\ell) &= z^2 \ell^{-2-d} c \\ c'_4(\ell) &= z^2 \ell^{-4-d} c_4 \\ h'(\ell) &= zh. \end{cases} \quad (82)$$

Wilson's Momentum Shell RG: Gaussian model

Imposing again $c' = c$ by setting $z = \ell^{1+d/2}$, we have

$$\begin{cases} t'(\ell) &= \ell^2 t \\ c'_4(\ell) &= \ell^{-2} c_4 \\ h'(\ell) &= \ell^{1+d/2} h \end{cases} \quad (83)$$

with fixed point $t^* = h^* = c_4^* = 0$, known as the **Gaussian fixed point**.

The *anomalous scaling dimensions* $\{y_i\}$ at the Gaussian f.p. are

$$y_t = 2, \quad y_{c_4} = -2, \quad y_h = 1 + d/2 \quad (84)$$

i.e. h is relevant while the higher-order transport term is not.

From previous scaling analysis, we can compute the critical exponents, e.g.

$$\nu_G = \frac{1}{y_t} = \frac{1}{2} \quad (85)$$

$$\Delta = \frac{y_h}{y_t} = \frac{1}{2} + \frac{d}{4} \quad (86)$$

Wilson's Momentum Shell RG: Gaussian model

Then, applying the Josephson's hyperscaling relation, we also find

$$\alpha_G = 2 - d\nu_G = 2 - \frac{d}{2} \quad (87)$$

Since also $\eta_G = 0$, we also obtain $\gamma_G = (2 - \eta_G)\nu_G = 1$.

Finally, we have

$$\beta_G = \frac{d - y_h}{y_t} = \frac{d - 2}{4} \quad (88)$$

and using the relation $\beta\delta = \Delta$, we find

$$\delta_G = \frac{\Delta}{\beta_G} = \frac{y_h/y_t}{(d - y_h)/y_t} = \frac{d + 2}{d - 2}. \quad (89)$$

REMARK: exponents γ_G, η_G, ν_G are the MF values, while $\alpha_G, \beta_G, \delta_G$ are different ($\alpha_{MF} = 0$, $\beta_{MF} = 1/2$, $\delta_{MF} = 3$).

Wilson's Momentum Shell RG: Gaussian model

REMARK: Can we systematically determine if terms

$$g_n \int d\vec{x} \phi^n(\vec{x}), \quad (90)$$

are relevant at the Gaussian fixed point?

Applying RG, i.e. $\vec{x}' = \vec{x}/\ell$ and $\phi'(\vec{x}') = \zeta^{-1}\phi(\vec{x})$ and imposing the same functional form [Notice that $z = \ell^{1+d/2}$ gives $\zeta = \ell^{1-d/2}$],

$$g_n \int d\vec{x}' \ell^d \ell^{-\frac{dn}{2}+n} \phi'(\vec{x}')^n = g'_n \int d\vec{x}' \phi(\vec{x}')^n, \quad (91)$$

we obtain the RG flow equation

$$g'_n(\ell) = \ell^{n-d(\frac{n}{2}-1)} g_n \quad \implies \quad y_n = n - d \left(\frac{n}{2} - 1 \right) \quad (92)$$

In particular, $O(\phi^4)$ terms irrelevant for $d > 4$, $O(\phi^6)$ terms for $d > 3$.

Wilson's Momentum Shell RG: Gaussian model

REMARK: If the $O(\phi^4)$ term is irrelevant for $d > 4$, why scaling analysis of the Gaussian model agrees with MF only for $d = 4$?

In the MF analysis, $M_0 \sim (rt/u)^\beta$, under rescaling we have

$$t' \sim \ell^{y_t} t, \quad M' \sim \ell^{y_M} M, \quad u' \sim \ell^{y_u} u \quad (93)$$

which gives

$$M'_0 \ell^{-y_M} \sim \left(\frac{rt' \ell^{-y_t}}{u' \ell^{-y_u}} \right)^\beta \implies y_M = \beta (y_t - y_u) \quad (94)$$

Using previous results $y_t = 2$, and $y_u = 4 - d$ and $y_M = -\zeta = d/2 - 1$

$$\beta = \frac{d - 2}{2(d - 2)} = \frac{1}{2}, \quad \forall d \geq 4 \quad (95)$$

The MF exponent is correct for $d \geq 4$ and the $O(\phi^4)$ term is a ***dangerously irrelevant*** term.